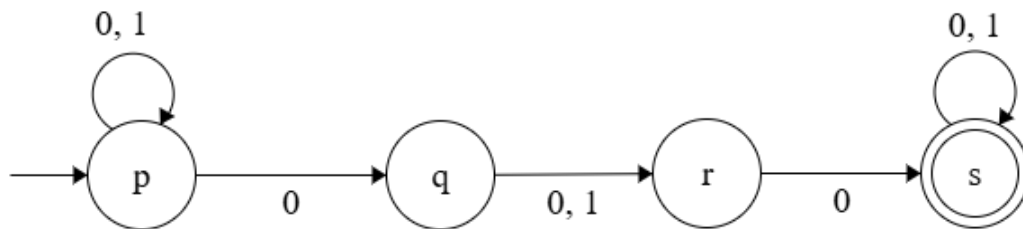


Oefeningen talen en automaten - Reeks 3

(Verbeterde) eindversie oefeningenopdracht 2

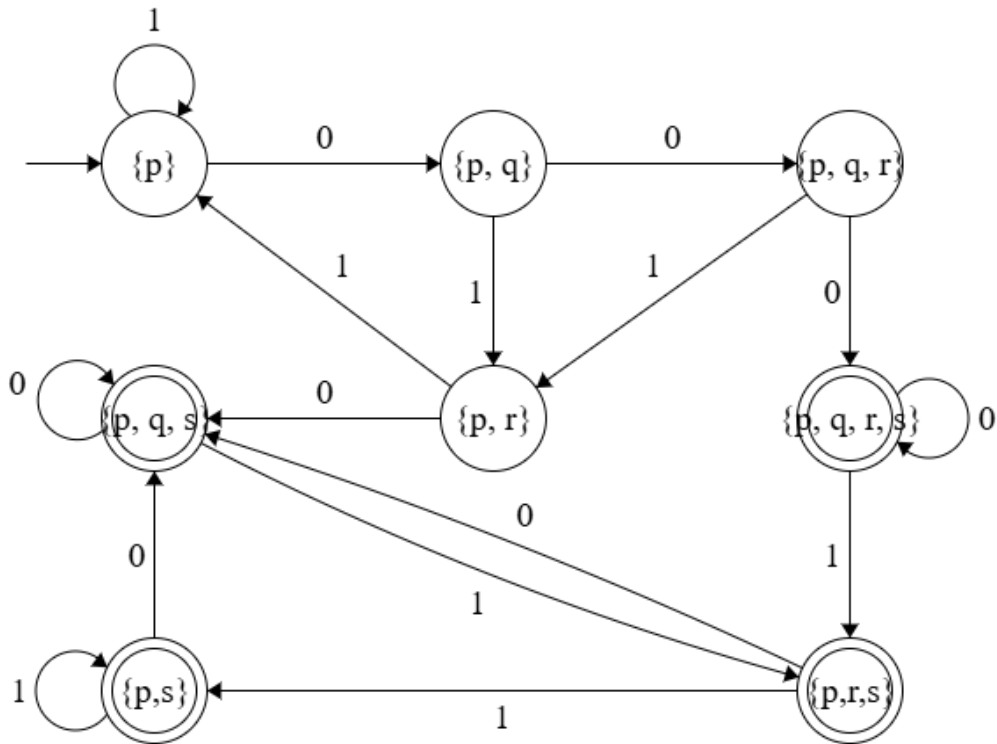
Reeks 3

1.



DFA

	0	1
$\rightarrow \{p\}$	$\{p, q\}$	$\{p\}$
$\{p, q\}$	$\{p, q, r\}$	$\{p, r\}$
$\{p, q, r\}$	$\{p, q, r, s\}$	$\{p, r\}$
$\{p, r\}$	$\{p, q, s\}$	$\{p\}$
$\ast\{p, q, r, s\}$	$\{p, q, r, s\}$	$\{p, r, s\}$
$\ast\{p, q, s\}$	$\{p, q, s\}$	$\{p, r, s\}$
$\ast\{p, r, s\}$	$\{p, q, s\}$	$\{p, s\}$
$\ast\{p, s\}$	$\{p, q, s\}$	$\{p, s\}$



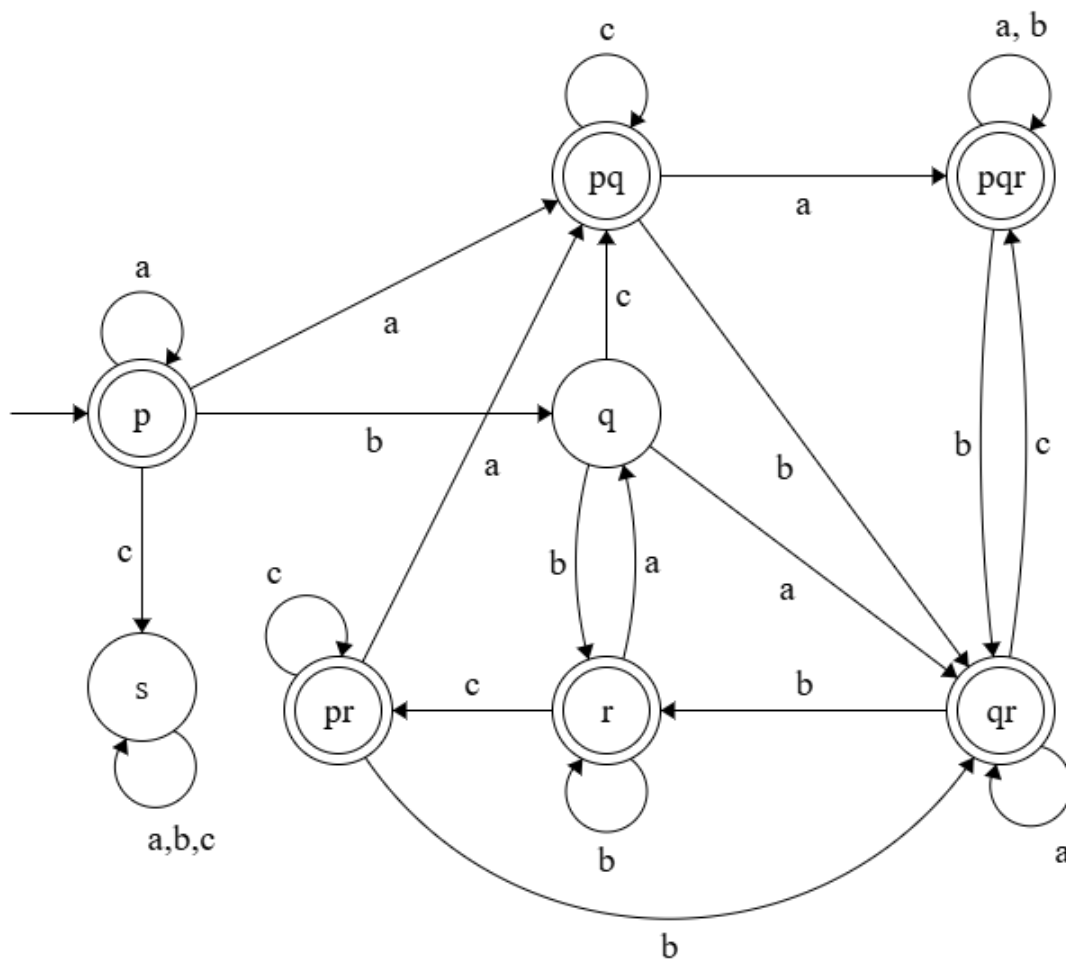
2.

DFA

Transitie Tabel

	a	b	c
→ *p	pq	q	s
*pq	pqr	qr	pq
q	qr	r	pq
s	s	s	s
*pqr	pqr	qr	pqr
*qr	qr	r	pqr
*r	q	r	pr
*pr	pq	qr	pr

Transitiedigram



3.

transitietabel die bij deze NFA hoort

	0	1
$\rightarrow *q0$	$\emptyset$	$\{q1, q\}$
$q1$	$\{q\}$	$\emptyset$
$*q$	$\emptyset$	$\emptyset$

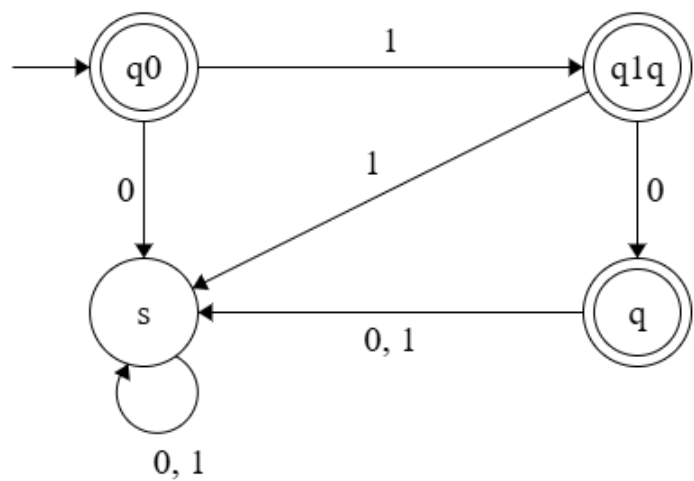
DFA

$D = \{\{q0, q1q, q, s\}, \{0, 1\}, \{q0, q1q, q, s\} \times \{0, 1\} \rightarrow \{q0, q1q, q, s\}, \{q0\}, \{q0, q1q, q\}\}$

Transitietabel

	0	1
→ *q0	s	q1q
*q1q	q	s
*q	s	s
s	s	s

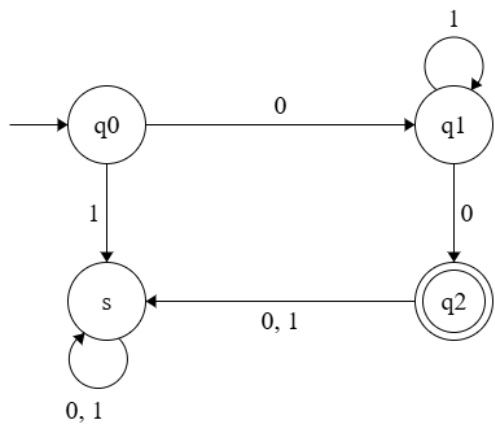
Transitiedigram



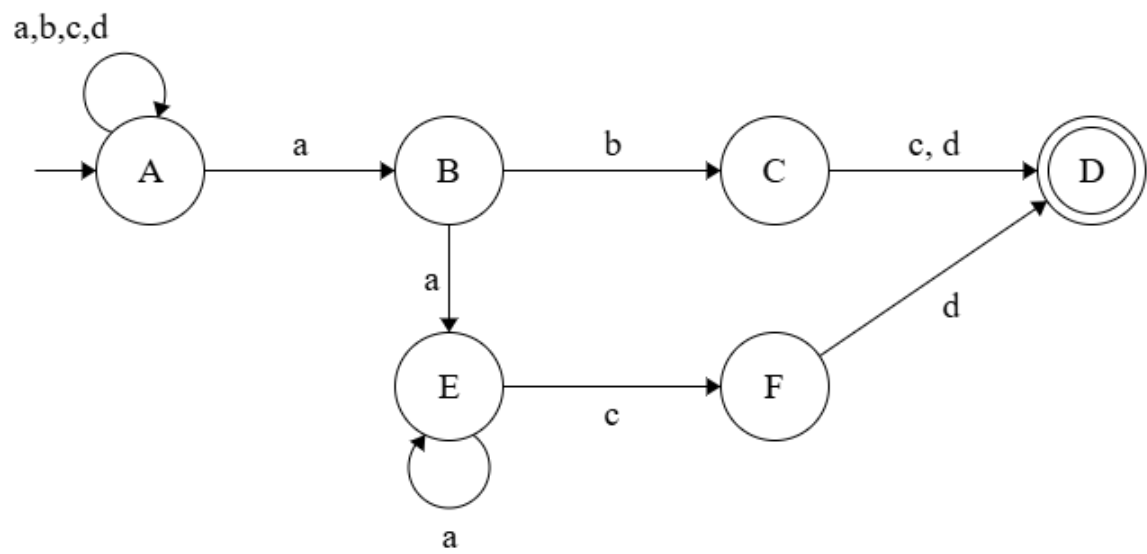
4.
 Het diagram stelt een NFA, want niet alle symbolen worden gebruikt bij ieder toestand.

DFA

	0	1
→ q0	q1	s
q1	q2	q1
*q2	s	s
s	s	s



5.

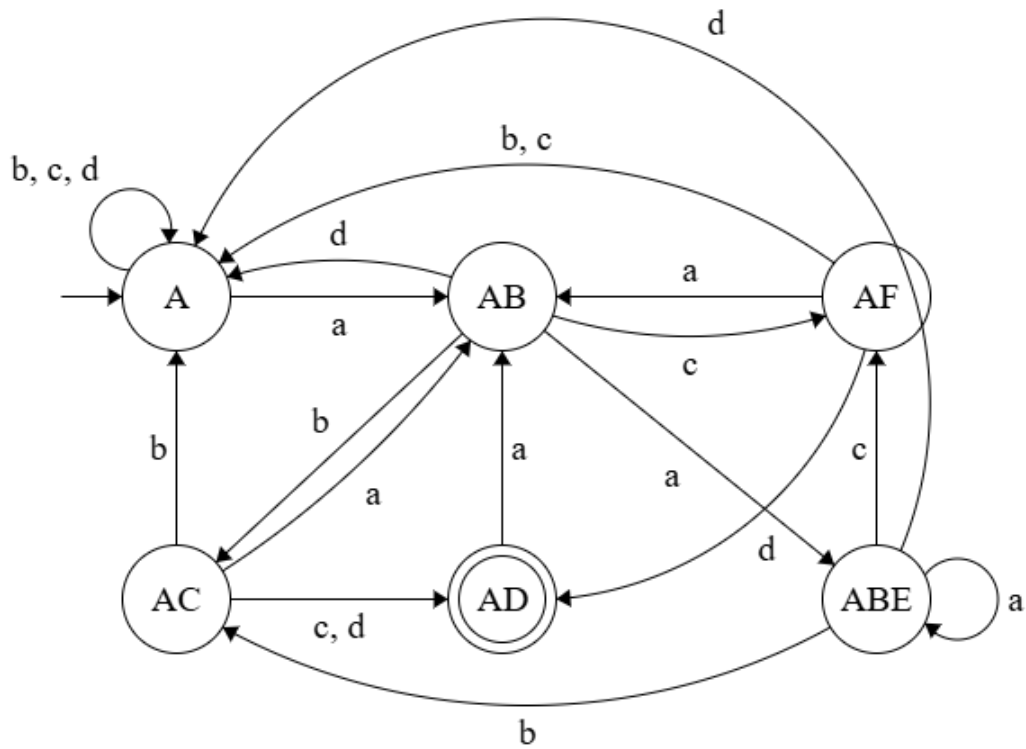


	A	B	C	D
→ A	{A, B}	{A}	{A}	{A}
B	{E}	{C}	∅	∅
C	∅	∅	{D}	{D}
*D	∅	∅	∅	∅
E	{E}	∅	{F}	∅
F	∅	∅	∅	{D}

6.

DFA

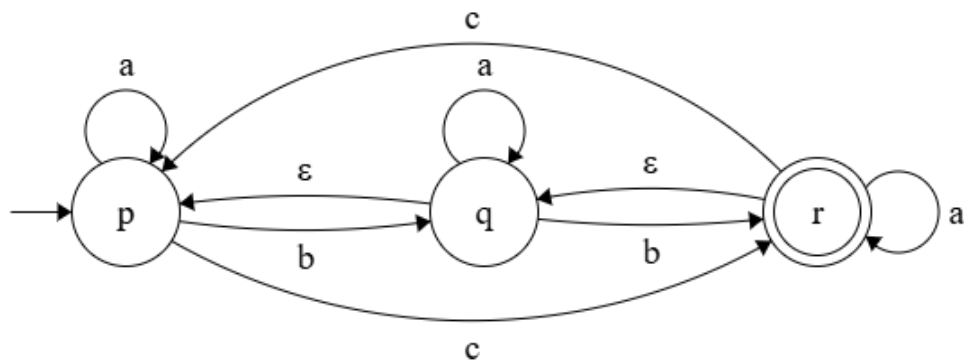
	A	B	C	D
→ A	AB	A	A	A
AB	ABE	AC	A	A
AC	AB	A	AD	AD
*AD	AB	A	A	A
AF	AB	A	A	AD
ABE	ABE	AC	AF	A



Reeks 4

1.

a.



b.

$$Eclosure(p) = \{p\}$$

$$Eclosure(q) = \{p, q\}$$

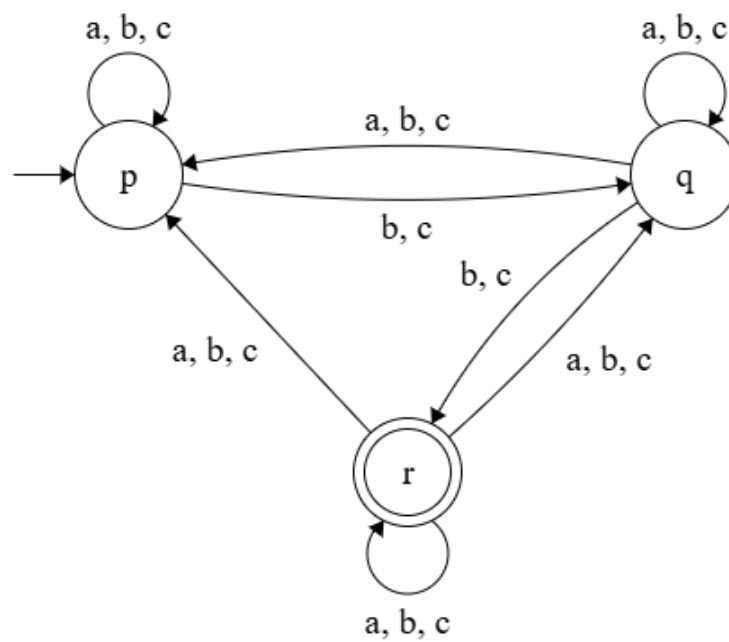
$$\text{Eclosure}(r) = \{p, q, r\}$$

c.

	ϵ^*	a	ϵ^*		ϵ^*	b	ϵ^*
p	$\{p\}$	$\{p\}$	$\{p\}$	p	$\{p\}$	$\{q\}$	$\{p, q\}$
q	$\{p, q\}$	$\{p, q\}$	$\{p, q\}$	q	$\{p, q\}$	$\{q, r\}$	$\{p, q, r\}$
r	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$	r	$\{p, q, r\}$	$\{q, r\}$	$\{p, q, r\}$
	ϵ^*		c	ϵ^*			
p	$\{p\}$		$\{r\}$	$\{p, q, r\}$			
q	$\{p, q\}$		$\{r\}$	$\{p, q, r\}$			
r	$\{p, q, r\}$		$\{p, r\}$	$\{p, q, r\}$			

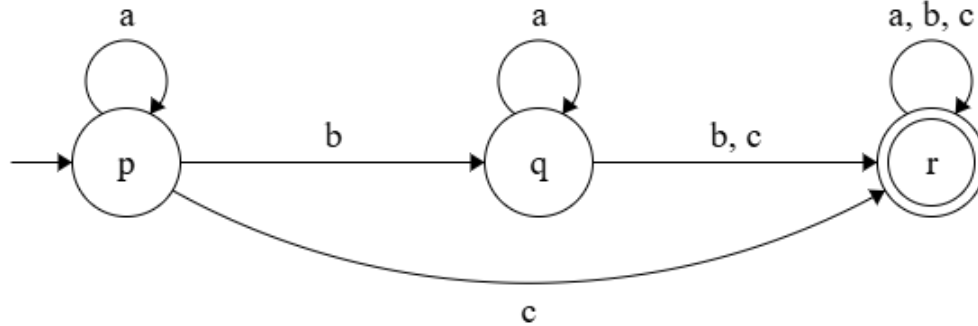
NFA

	A	B	C
$\rightarrow P$	$\{p\}$	$\{p, q\}$	$\{p, q, r\}$
Q	$\{p, q\}$	$\{p, q, r\}$	$\{p, q, r\}$
*R	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$



DFA

	A	B	C
$\rightarrow P$	p	q	r
Q	q	r	r
*R	r	r	r



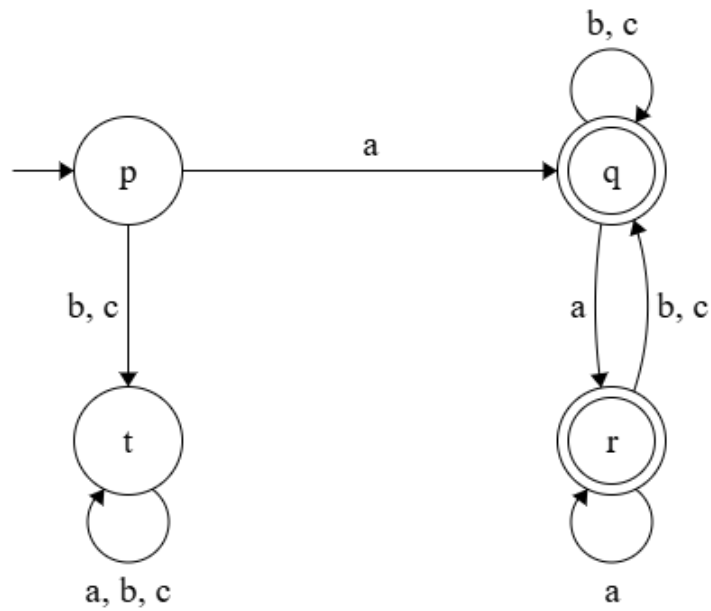
2.

e-NFA

	ϵ	a	b	c
$\rightarrow p$	{s, r}	{p, q}	$\emptyset$	$\emptyset$
*q	{r}	$\emptyset$	{p, q}	$\emptyset$
r	$\emptyset$	{r, s}	$\emptyset$	{q, r}
s	{r}	$\emptyset$	$\emptyset$	{s}

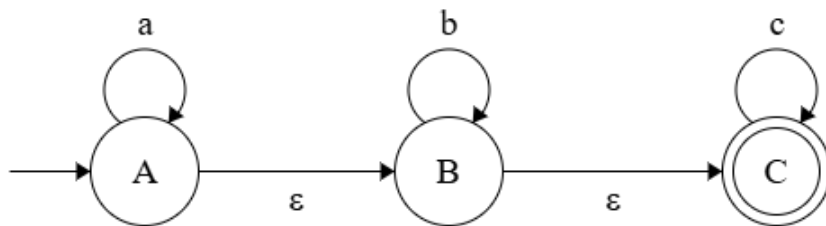
DFA

	a	b	c
$\rightarrow \{p, s, r\}$	{p, q, r}	$\emptyset$	{r, s}
*{p, q, r}	{p, q, r, s}	{p, q, r}	{q, r}
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
*{p, q, r, s}	{p, q, r, s}	{p, q, r}	{q, r, s}
*{q, r}	{r, s}	{p, q, r}	{q, r}
*{q, r, s}	{r, s}	{p, q, r}	{q, r, s}
{r, s}	{r, s}	$\emptyset$	{q, r, s}



3.

Transitiedigram



Transitietabel

	ϵ	a	b	c
$\rightarrow A$	{A, B, C}	{A}	$\emptyset$	$\emptyset$
B	{B, C}	$\emptyset$	{B}	$\emptyset$
*C	{C}	$\emptyset$	$\emptyset$	{C}

4.

DFA

Transitietabel

	a	b	c
$\rightarrow *{A, B, C}$	{A, B, C}	{B, C}	{C}
*{B, C}	$\emptyset$	{B, C}	{C}

$\ast\{C\}$	$\emptyset$	$\emptyset$	$\{C\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Transitiediagram

